

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

May 2026

CP3 - Communication Practice

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) In Country G, SNG is a defined benefit pension scheme established by Company A for its employees. The scheme is managed by its Trustees who make sure that the scheme rules are followed. You are the Chief Financial Officer (CFO) and a qualified Actuary at Insurance Company R – a life insurance company that has been selling individual annuity products in Country G for the last 15 years. Insurance Company R has not previously entered the group annuity market.

The Board of Insurance Company R is considering entering the group annuity business by providing a bulk annuity buyout to the trustees and sponsor of SNG. If this buyout is successfully completed, it will be Insurance Company R's first ever group annuity contract.

As the CFO and a qualified Actuary, you have been asked by the Board to prepare a paper for the upcoming Board meeting to help the Board evaluate this strategic opportunity. The Board comprises the Managing Director, the Chief Risk Officer, the Chief Investment Officer, the Appointed Actuary, and three Independent Non-Executive Directors, not all of whom have an actuarial background.

The most recent actuarial valuation of SNG (as at 1 April 2025) has revealed the following position:

	1 April 2024 (₹ Crore)	1 April 2025 (₹ Crore)
Market value of scheme assets	3,840	4,080
Actuarial value of liabilities (Technical Provisions)	4,960	4,720
Funding level	77%	86%
Annual employer contribution under recovery plan (₹ Crore)	144	144

Insurance Company R has quoted a buyout premium of ₹5,200 crore to take over full responsibility for paying all pension benefits to SNG's members. Company A will need to fund the shortfall between the buyout premium and current scheme assets by making an additional one-off contribution to SNG.

As the CFO of Insurance Company R, draft a paper (in approximately 900 words) for presentation at the upcoming Board meeting to evaluate the proposed group annuity buyout of SNG. The Board comprises members with varying levels of actuarial expertise.

The paper should cover the following:

- Background to SNG and why Company A is seeking a buyout.
- The key financial implications for Insurance Company R of writing this new group annuity contract, including the premium receivable and the one-time costs of entering a new line of business.
- The main advantages and risks for Insurance Company R of entering the group annuity market for the first time.
- The key risks to scheme members and Company A, and how these are mitigated under the buyout.

- Your recommendation to the Board on whether Insurance Company R should proceed with this contract.

[90 Marks]

Q.2) Explain how you decided on:

i) The depth of your explanation [5]

ii) Filtering information [5]

[10 Marks]

[The information given below is fictitious and does not belong to any company or individual. It is meant to be used only for the purpose of this examination.]

Annexure:

Section 1: Key Definitions and Background

Defined Benefit Pension Scheme

A defined benefit (DB) pension scheme is an occupational retirement arrangement in which the employer promises to pay each employee a pension at retirement that is calculated using a pre-defined formula – typically based on the employee's final salary (or career average salary) and the number of years of service completed. Unlike a defined contribution scheme, where the retirement benefit depends on how much has been contributed and how investments have performed, a DB scheme guarantees a specific income to the member for life, regardless of investment returns. The financial risk of meeting these guaranteed benefits lies entirely with the sponsoring employer.

In Country G, DB schemes established by private sector employers operate as registered trust-based arrangements, governed by a board of trustees who are responsible for managing the scheme's assets and ensuring that pensions are paid to members. Employers (sponsors) are legally obligated to fund any shortfall between the scheme's assets and its liabilities.

Group Annuity Buyout

A group annuity buyout (also known as a pension scheme buyout or bulk annuity buyout) is a transaction in which a life insurance company takes on full legal responsibility for paying pension benefits to all members of a DB scheme, in exchange for a single premium paid by the trustees and the sponsoring employer. Once the buyout is complete:

- Each member receives an individual annuity policy issued directly in their name by the insurer.
- The sponsoring employer (Company A) has no further legal obligation in relation to the pension scheme.
- The pension scheme is wound up and ceases to exist as a trust.

A buyout differs from a buy-in (where the insurance policy is held by the trustees as a scheme asset, and the scheme continues) in that a buyout fully extinguishes the employer's pension liability. It is an irrevocable transaction.

Individual Annuity Business (Insurance Company R's existing business)

An individual annuity is a contract between a single policyholder and a life insurance company. The policyholder pays a premium and in return receives a guaranteed income for life (or for a specified period). Insurance Company R has been selling individual annuity products in Country G for 15 years and has a well-established pricing, underwriting and administration capability for individual risks. Group annuity business differs in that the insurer prices and manages a large, heterogeneous pool of members as a single transaction, requiring different actuarial models, asset-liability management frameworks and administrative capabilities.

Section 2: Background to SNG and Company A

SNG is a defined benefit pension scheme that has been in operation for over 25 years. It was established by Company A – a large manufacturing conglomerate in Country G – to provide pension security for its long-serving employees. The scheme provides a pension at retirement equal to 1/50th of the member's final salary for each year of pensionable service, with pension increases in payment linked to inflation, capped at 4% per annum.

The scheme was closed to new entrants in 2018, when Company A introduced a defined contribution scheme for all new employees. However, existing members continue to accrue benefits. The scheme membership as at 1 April 2025 is approximately:

- Active members (still employed and accruing benefits): 1,200
- Deferred members (left employment, benefits preserved): 2,100
- Pensioners (currently receiving pension): 900

Total membership: approximately 4,200 members.

Company A has faced significant financial pressure over recent years due to a downturn in the manufacturing sector, rising input costs and increased global competition. Despite these challenges, Company A has continued to make annual contributions of ₹144 crore to SNG under an agreed recovery plan. Company A's Board has become increasingly concerned that the pension scheme represents an open-ended financial liability that could threaten the viability of the business in the event of adverse investment returns or members living longer than expected.

Section 3: Current Funding Position of SNG

The most recent actuarial valuation of SNG, completed as at 1 April 2025, shows the following position:

	1 April 2024 (₹ Crore)	1 April 2025 (₹ Crore)
Market value of scheme assets	3,840	4,080
Technical provisions (actuarial liabilities)	4,960	4,720
Funding deficit	1,120	640
Funding level	77%	86%
Annual recovery plan contribution (₹ Crore)	144	144

Note: Technical provisions are the actuarial estimate of the assets required today to meet all future pension payments to members, calculated using assumptions agreed between the

Scheme Actuary and the trustees. The key assumptions used in the 1 April 2025 valuation are:

- Discount rate: 6.80% per annum (based on expected long-term return on scheme assets, which are invested in a mix of government bonds and equity funds)
- Inflation assumption: 4.50% per annum (for both pension increases and salary increases)
- Mortality: LIC (2006-08) Ultimate mortality tables with an improvement factor of 1.5% per annum
- Retirement age: 60 years

The valuation shows that the scheme's funding position has improved from 77% (deficit of ₹1,120 crore) in 2024 to 86% (deficit of ₹640 crore) in 2025, reflecting investment returns above expectations and ongoing employer contributions. However, a material deficit of ₹640 crore remains.

Section 4: Buyout Proposal – Details and Financial Analysis

Insurance Company R has submitted a quotation to provide a full buyout of SNG. The key terms of the proposal are:

Item	Amount (₹ Crore)
Buyout premium quoted by Insurance Company R	5,200
Current scheme assets to be transferred on completion	4,080
Shortfall – additional contribution required from Company A	1,120
Estimated one-time system and operational set-up costs for Insurance Company R	180
Expected annual group annuity outgo (pensions payable to members)	~520

The buyout premium of ₹5,200 crore is higher than the technical provisions of ₹4,720 crore. This is because Insurance Company R prices the buyout using a more prudent, risk-free discount rate (based on Government of Country G bond yields of approximately 6.20% per annum) rather than the scheme's asset-return-based discount rate. Additionally, Insurance Company R applies more conservative mortality assumptions to reflect the fact that it is taking on a guaranteed liability for the lifetime of all members. The difference of approximately ₹480 crore above technical provisions represents the cost of risk transfer.

Note: The buyout premium of ₹5,200 crore includes, in addition to the cost of securing all member benefits priced on a risk-free basis, a loading to recover Insurance Company R's one-time set-up costs and a profit margin.

Company A's financial position (as at 31 March 2025):

- Net assets: ₹2,560 crore
- Annual revenue: ₹3,200 crore
- Net borrowings: ₹760 crore
- Annual operating cash flow: ₹380 crore
- Current recovery plan contribution to SNG: ₹144 crore per annum (committed until 2032)

Company A has indicated it would need to raise external financing to fund the additional contribution of ₹1,120 crore required to complete the buyout transaction. However, if the buyout proceeds, Company A's annual contributions of ₹144 crore to SNG would cease entirely.

Section 5: Insurance Company R – Business Context

Insurance Company R is one of the leading individual annuity providers in Country G, with a 15-year track record of managing longevity, interest rate and expense risks in its individual annuity book. It currently manages an individual annuity portfolio with reserves of approximately ₹18,000 crore and has an internal solvency ratio of 165% (against a regulatory minimum of 150% prescribed by Country G's Insurance Regulator).

The Appointed Actuary of Insurance Company R has carried out a preliminary assessment of the impact of writing this group annuity contract on the company's solvency position. The key findings are:

- The buyout contract, once written, will increase Insurance Company R's total annuity reserves by approximately ₹5,200 crore.
- The Appointed Actuary estimates that, under the base scenario (where pricing assumptions are met), the solvency ratio of Insurance Company R post-transaction will be approximately 156% – above the regulatory minimum of 150%.
- Under a stressed scenario (where longevity is 10% worse than assumed and investment returns are 50 basis points lower than expected), the estimated solvency ratio falls to approximately 151% – still above the regulatory minimum, though with limited headroom.
- The Appointed Actuary has recommended that, to provide additional protection against the stressed scenario, Insurance Company R should arrange longevity reinsurance for this contract prior to completion. The estimated cost of this reinsurance cover is ₹60 crore, which is not included in the one-time set-up cost of ₹180 crore quoted above.

Insurance Company R has never previously written a group annuity contract. The key differences between individual and group annuity business that the Board must understand are:

Feature	Individual Annuity	Group Annuity (Buyout)
Premium size	Small – typically ₹5 lakh to ₹2 crore per policy	Very large – ₹5,200 crore in a single transaction
Pricing basis	Individual health and mortality assessment	Pooled assessment of large member cohort
Administration	Standardised individual policy administration	Complex – must administer multiple member categories (active, deferred, pensioner) with varying benefit structures
Asset-Liability Management (ALM)	Well-established individual annuity ALM framework	New ALM framework required for long-duration group liabilities
Reinsurance	Individual per-policy reinsurance in place	A separate portfolio-level reinsurance arrangement is required, covering the risk that the entire member cohort lives longer than assumed (different

		from per-policy individual reinsurance)
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Insurance Company R's Board is aware that group annuity business is a growing segment in Country G as many large corporates look to de-risk their DB pension obligations. The total addressable market for group annuity buyouts in Country G is estimated to be in excess of ₹80,000 crore over the next 10 years, as private sector DB schemes seek to transfer liabilities to insurers.

The Board of Insurance Company R has previously discussed the possibility of entering the group annuity market and has expressed a broad interest in doing so, subject to the risks being manageable and the transaction being financially sound. The SNG buyout has been identified internally as a suitable first transaction because:

- SNG's membership data has been audited by an independent administrator and is considered to be of good quality, reducing data risk.
- Company A has confirmed its commitment to fund the ₹1,120 crore additional contribution and has received indicative financing terms from its bankers.
- The scheme has a relatively straightforward benefit structure (final salary, with a single inflation-linked increase cap), making it more tractable to price and administer than schemes with complex discretionary benefit histories.

The Board has asked the CFO to present a clear recommendation, with supporting rationale, at the upcoming Board meeting.

Section 6: Key Risks and Alternative Options

The following key risks have been identified for both Insurance Company R and scheme members/Company A:

Risks for Insurance Company R:

- Longevity risk: Members living longer than assumed in pricing, leading to higher-than-expected annuity payments.
- Pricing risk: If the buyout premium has been underpriced due to incorrect assumptions (mortality, inflation, expenses), Insurance Company R will incur losses on the contract.
- Operational risk: First-time entry into group annuity administration carries significant operational risk, including data migration errors, benefit calculation errors and regulatory non-compliance.
- Asset-Liability Management (ALM) risk: The group annuity liabilities are very long-duration (some members may be under 40 years old as active members). Matching these liabilities requires long-duration assets (government bonds), which may be in short supply in Country G's market.
- Concentration risk: A single contract of ₹5,200 crore represents approximately 29% of Insurance Company R's existing reserves. This is a significant concentration of risk in a single transaction.
- Reputational risk: A failure to administer the group annuity contract correctly could damage Insurance Company R's brand and its existing individual annuity business.

Risks for Company A and SNG Members:

- Counterparty risk: Once the buyout is complete and the scheme is wound up, members depend entirely on Insurance Company R to pay their pensions. If Insurance Company R were to fail, members may receive reduced benefits. (Note: Country G's Insurance Regulator requires insurers to hold capital in excess of their liabilities, which provides a buffer.)
- Regulatory protection: Under the current trust-based arrangement, scheme assets are ring-fenced and protected from Company A's insolvency. Post-buyout, members' pensions are secured by Insurance Company R's financial strength and regulatory capital requirements.
- Benefit flexibility: Once benefits are secured under the buyout, members lose any discretionary benefit improvements that trustees might otherwise have been able to grant (e.g. above-inflation increases in good years).

Alternative Options available to Company A and the Trustees:

- Buy-in only (partial risk transfer): A bulk annuity policy covering pensioners only (approximately ₹1,560 crore of liability) could be purchased, leaving active and deferred liabilities within the scheme. This would require a smaller upfront additional contribution from Company A.
- Longevity swap: A financial instrument that transfers only longevity risk (the risk of members living longer than expected) to a bank or reinsurer. Assets remain within the scheme, and Company A retains investment and inflation risk.
- Liability-Driven Investment (LDI) strategy: The scheme could adopt an LDI approach, investing predominantly in bonds to match liabilities, reducing funding volatility and continuing on a self-sufficiency basis. Annual contributions continue until the scheme reaches full funding.
